

3. A VARIATIONAL CHARACTERIZATION OF THE ENTROPY DENSITY

Theorem 3.1. *For any possible pair (e, t) , $s_{e,t} = -\min I(g)$, where the minimum is over all graphons g with $e(g) = e$ and $t(g) = t$, where*

$$e(g) = \int_0^1 \int_0^1 g(x, y) dx dy, \quad t(g) = \int_0^1 \int_0^1 \int_0^1 g(x, y) g(y, z) g(z, x) dx dy dz,$$

and the rate function is

$$(8) \quad I(g) = \int_0^1 \int_0^1 I_0(g(x, y)) dx dy, \text{ where } I_0(u) = \frac{1}{2} [u \ln(u) + (1 - u) \ln(1 - u)].$$

Proof. We first prove that $s_{e,t}$ is well-defined. A priori all we know is that $\liminf \ln(Z_{e,t}^{n,\delta})/n^2$ and $\limsup \ln(Z_{e,t}^{n,\delta})/n^2$ exist as $n \rightarrow \infty$. However, we will show that they both approach $-\min I(g)$ as $\delta \rightarrow 0^+$.

We need to define a few sets. Let U_δ be the set of graphons g with $e(g)$ and $t(g)$ strictly within δ of e and t , i.e. the preimage of an open square of side 2δ in (e, t) -space, and let F_δ be the preimage of the closed square. Let $\tilde{U}_\delta$ and $\tilde{F}_\delta$ be the corresponding sets in $\tilde{\mathcal{W}}$. Let $|U_\delta^n|$ and $|F_\delta^n|$ denote the number of graphs with n vertices whose checkerboard graphons (4) lie in U_δ or F_δ . The large deviation principle, Theorem 2.3 of [CV], implies that:

$$(9) \quad \limsup_{n \rightarrow \infty} \frac{\ln |F_\delta^n|}{n^2} \leq - \inf_{\tilde{g} \in \tilde{F}_\delta} I(\tilde{g}),$$

which also equals $-\inf_{g \in F_\delta} I(g)$, and that

$$(10) \quad \liminf_{n \rightarrow \infty} \frac{\ln |U_\delta^n|}{n^2} \geq - \inf_{\tilde{g} \in \tilde{U}_\delta} I(\tilde{g}),$$

which also equals $-\inf_{g \in U_\delta} I(g)$. This yields a chain of inequalities

$$(11) \quad - \inf_{U_\delta} I(g) \leq \liminf \frac{\ln |U_\delta^n|}{n^2} \leq \limsup \frac{\ln |U_\delta^n|}{n^2} \leq \limsup \frac{\ln |F_\delta^n|}{n^2} \leq - \inf_{F_\delta} I(g) \leq - \inf_{U_{\delta+\delta^2}} I(g)$$

As $\delta \rightarrow 0^+$, the limits of $-\inf_{U_\delta} I(g)$ and $-\inf_{U_{\delta+\delta^2}} I(g)$ are the same, and everything in between is trapped.

So far we have proven that

$$(12) \quad s_{e,t} = - \lim_{\delta \rightarrow 0^+} \inf_{U_\delta} I(g).$$

Next we must show that the right hand side is equal to $-\min_{F_0} I(g)$. By definition, we can find a sequence of reduced graphons $\tilde{g}_\delta \in \tilde{U}_\delta$ such that $\lim_{\delta \rightarrow 0} I(\tilde{g}_\delta) = \liminf_{U_\delta} I(g)$. Since $\tilde{\mathcal{W}}$ is compact, these reduced graphons converge to a reduced graphon $\tilde{g}_0$, represented by a graphon $g_0 \in F_0$. Since I is lower-semicontinuous [CV], $I(g_0) \leq \liminf_{F_0} I(g)$, so $\min_{F_0} I(g) \leq \liminf_{U_\delta} I(g)$. (We write min rather than inf since $\tilde{F}_0$ is compact.) However, $\min_{F_0} I(g)$ is at least as big as $\inf_{U_\delta} I(g)$, since $F_0 \subset U_\delta$. Thus $\min_{F_0} I(g) = \liminf_{\delta \rightarrow 0} \inf_{U_\delta} I(g)$. $\square$